



FORM2 - FINAL EXAM

Questions (Scorlar)

An entity has :() A set of properties(ii) A set of properties and values for all the properties(iii) A set of properties and the values for some set of properties may non-uniquely identify an entity(iv) A set of properties and the values for some set of properties may uniquely identify an entityWhich of the above,(i, ii,iii,iv) are valid ?

- ☐ (ii) only
- ☐ (iii) only
- ☐ (i) only
- ☒ (iv) only

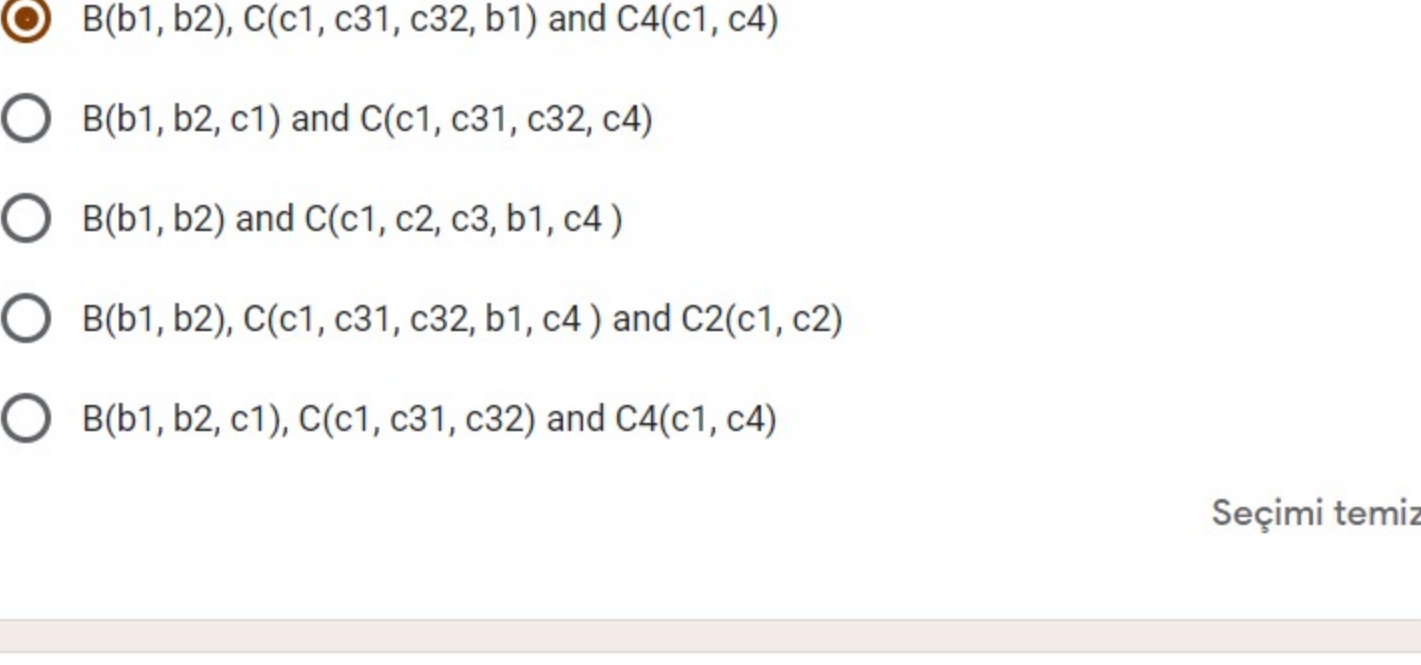
Seçimi temizle

In RDBMS, the constraint that no key attribute (column) may be NULL is referred to as:

- ☐ Multi-valued dependency
- ☐ Functional dependency
- ☒ Referential integrity
- ☐ Entity integrity

Seçimi temizle

Diagram of ER Model is given below, which option is true to describe the schema of Relational Model applying the converting steps for each feature.



- ☒ B(b1, b2), C(c1, c31, c32, b1) and C4(c1, c4)
- ☐ B(b1, b2, c1) and C(c1, c31, c32, c4)
- ☐ B(b1, b2) and C(c1, c2, c3, b1, c4)
- ☐ B(b1, b2), C(c1, c31, c32, b1, c4) and C2(c1, c2)
- ☐ B(b1, b2, c1), C(c1, c31, c32) and C4(c1, c4)

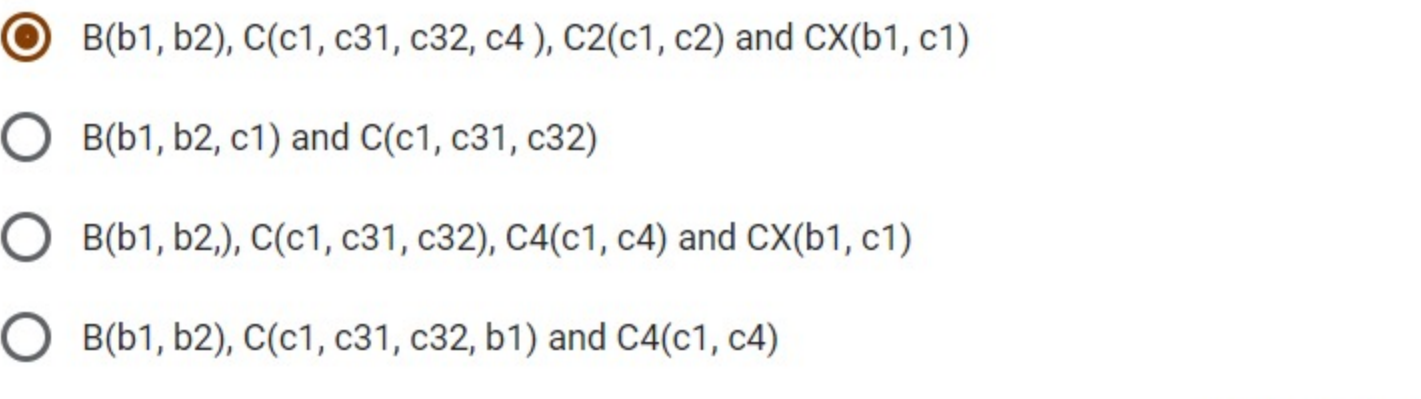
Seçimi temizle

Relational model differ from E/R model in that

- ☐ All tables are inherently equivalent to weak entities.
- ☐ The concept of multiway relationship cannot be expressed.
- ☒ Attributes / Columns are sometimes repeated between tables, unlike entities.
- ☐ The concept of relationship cannot be expressed.
- ☐ The concept of entity cannot be expressed.

Seçimi temizle

Diagram of ER Model is given below, which option is true to describe the schema of Relational Model applying the converting steps for each feature.



- ☐ B(b1, b2, c1), C(c1, c2, c3, b1, c4)
- ☒ B(b1, b2), C(c1, c31, c32, c4), C2(c1, c2) and CX(b1, c1)
- ☐ B(b1, b2, c1) and C(c1, c31, c32)
- ☐ B(b1, b2), C(c1, c31, c32), C4(c1, c4) and CX(b1, c1)
- ☐ B(b1, b2), C(c1, c31, c32, b1) and C4(c1, c4)

Seçimi temizle

Which action cannot be taken in case of integrity violation?

- ☐ None
- ☒ Specifying foreign keys for integrity constraints
- ☐ Trigger additional updates so the violation is corrected
- ☐ Cancel the operation that causes the violation
- ☐ Perform the operation but inform the user of the violation

Seçimi temizle

Relations in Relational Model produced from E-R Model will always be in

- ☐ Fourth normal form
- ☒ Third normal form
- ☐ Second normal form
- ☐ First normal form

Seçimi temizle

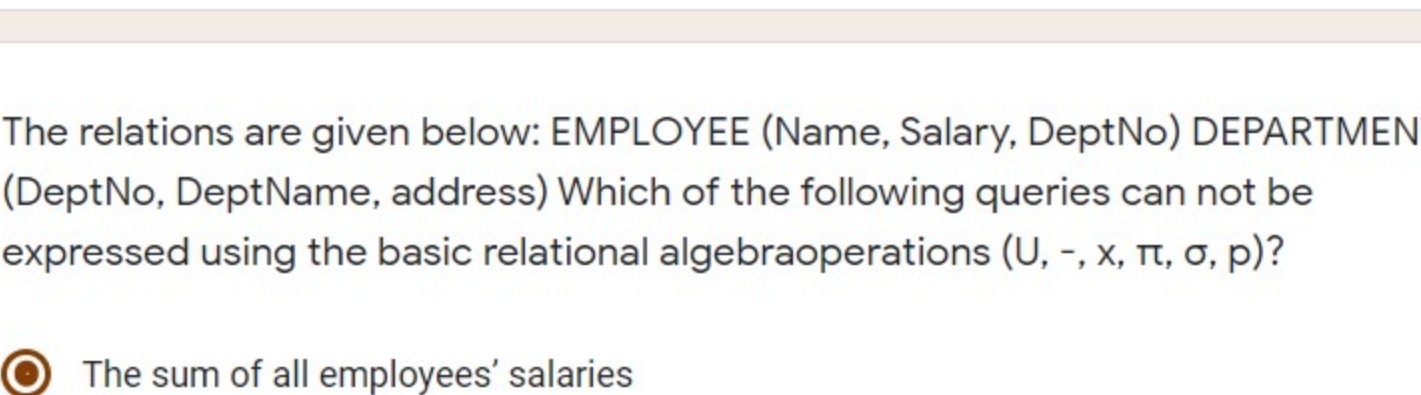
The following SCHOOL_DETAIL table is given, After applying the all normalisations, which option is the final result?

No#	SportType#	NumberofPeople	PricePerHour
1	SWIMMING	20	3
1	TENNIS	10	5
1	SQUASH	30	1
1	SAUNA	5	2
2	TENNIS	10	5
2	SQUASH	15	1
3	SWIMMING	10	3
3	TENNIS	20	5

- ☐ SCHOOL_DETAIL(No#, NumberofPeople), COST(SportType#, PricePerHour, No#)
- ☐ SCHOOL_DETAIL(No#, SportType#, COST(SportType#, NumberofPeople , PricePerHour)
- ☒ SCHOOL_DETAIL(No#, SportType#, COST(SportType#, PricePerHour), PERSON(SportType# NumberofPeople)
- ☐ SCHOOL_DETAIL(No#, SportType#, NumberofPeople), COST(SportType#, PricePerHour)

Seçimi temizle

Diagram of ER Model is given below, which option is true to describe the schema of Relational Model applying the converting steps for each feature.



- ☐ e) C(c1, c31, c32), C4(c1, c4), D(d1, d2) and Z(c1, d1)
- ☒ C(c1, c31, c32), D(c1, d1, d2) and C2(c1, c2)
- ☐ C(c1, c31, c32), C4(c1, c4) and D(c1, d1, d2)
- ☐ C(c1, d1, c31, c32), C4(c1, c4) and D(d1, d2)
- ☐ d) C(c1,c2, c3), D(c1, d1, d2) and C4(c1, c4)

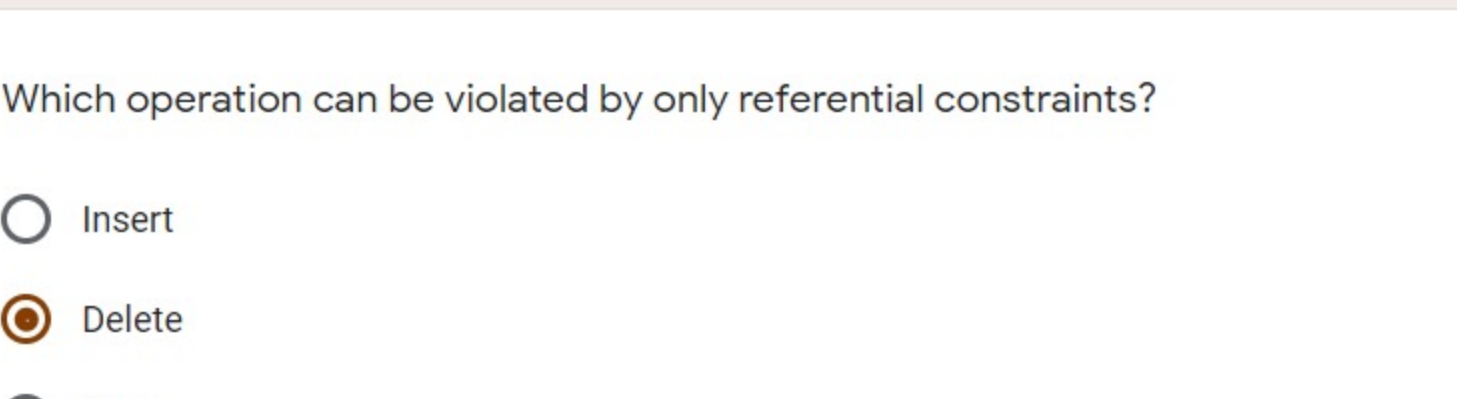
Seçimi temizle

The relations are given below: EMPLOYEE (Name, Salary, DeptNo) DEPARTMENT (DeptNo, DeptName, address) Which of the following queries can not be expressed using the basic relational algebraic operations (U, -, x, rt, o, p)?

- ☒ The sum of all employees' salaries
- ☐ DEPARTMENT (DeptNo, DeptName, address)
- ☐ Employees whose name is the same as their department name
- ☐ All employees of a given department

Seçimi temizle

Which option is not true after mapping the PRODUCT entity from ER Diagram to Relational Model below, considering Prdcode is numeric and PrdName is Varchar(50).



- ☐ PRODUCT (PrdCode#, PrdName, Color, NumberofProducts, Year, Season, Model)
- ☐ ALL
- ☒ PRODUCT (PrdCode#, PrdName#, Color, NumberofProducts, Type)
- ☐ PRODUCT (PrdCode#, PrdName, Color, NumberofProducts, Type)
- ☐ PRODUCT (PrdCode#, PrdName, Color, Year, Season, Model)

Seçimi temizle

Which operation can be violated by only referential constraints?

- ☐ Insert
- ☒ Delete
- ☐ None
- ☐ Create
- ☐ All

Seçimi temizle

Which option can be used while remedying by several actions in the case of possible violations?

- ☒ Restrict
- ☐ None
- ☐ Cascade
- ☐ All
- ☐ Set Null

Seçimi temizle

Based on the cardinality ratio and participation _____ associated with a relationship type, choose either the Foreign Key Design, the Cross Referencing Design or Mutual Referencing Design

- ☒ Constraints
- ☐ Keys
- ☐ Rules
- ☐ Entity
- ☐ Diger: _____

Seçimi temizle

The following SCHOOL_DETAIL table is given, which normalisation form should be applied first?

No#	SportType#	NumberofPeople	PricePerHour
1	SWIMMING	20	3
1	TENNIS	10	5
1	SQUASH	30	1
1	SAUNA	5	2
2	TENNIS	10	5
2	SQUASH	15	1
3	SWIMMING	10	3
3	TENNIS	20	5

- ☐ 3 NF
- ☐ 1NF
- ☒ 2NF
- ☐ BCNF

Seçimi temizle

When are the normalisation forms not needed for?

- ☒ Eliminate redundancy
- ☐ Remove duplicates
- ☐ Avoid anomalies
- ☐ Force integrity constraints

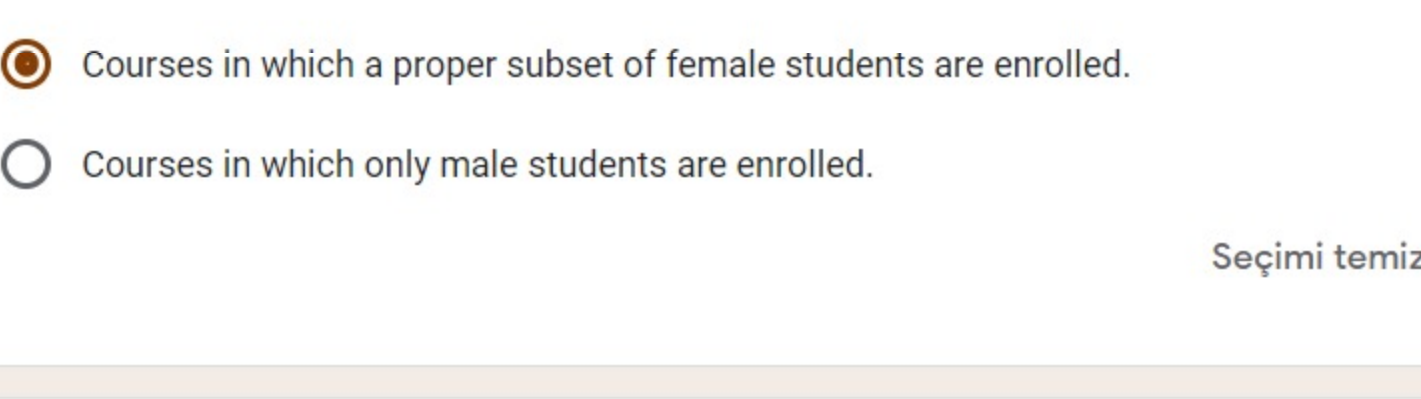
Seçimi temizle

Given 2 union-compatible tables M and N with 100 rows and 20 rows respectively. The minimum number of rows in (M – (M – N)) is 20.

- ☐ False
- ☒ True

Seçimi temizle

Diagram of ER Model is given below, which option is true to describe the schema of Relational Model applying the converting steps for each feature.



- ☐ A(a1, a2, b1, x1) and B(b1, b2)
- ☒ A(a1, a2) and B(b1, b2, a1, a2, x1)
- ☐ A(a1, a2) and B(b1, b2, x1)
- ☐ A(a1, a2) and B(b1, b2, a1, a2, x1)
- ☐ A(a1, a2, , x1) and B(b1, b2)

Seçimi temizle

Information about a collection of students is given by the relation studinfo(studid, name, sex). The relation enroll(studid, courseid) gives which student has enrolled for (or taken) that course(s). Assume that every course is taken by at least one male and at least one female student. What does the following relational algebra expression represent?

- $\Pi_{courseid} ((\Pi_{studid} (\sigma_{sex='female'} (studinfo)) \times \Pi_{courseid} (enroll)) - enroll)$
- ☐ None of the above
- ☐ Courses in which all the female students are enrolled.
- ☒ Courses in which a proper subset of female students are enrolled.
- ☐ Courses in which only male students are enrolled.

Seçimi temizle

Let R and S be two relations with the following schema R (P,Q,R1,R2,R3) and S (P,Q,S1,S2)Where (P, Q) is the key for both schemas. Which of the following queries, (I,II,III,IV) are equivalent?

- I. $\Pi_P (R \bowtie S)$
- II. $\Pi_P (R) \bowtie \Pi_P (S)$
- III. $\Pi_P ((\Pi_{P,Q} (R) \cap \Pi_{P,Q} (S)))$
- IV. $\Pi_P (\Pi_{P,Q} (R) - (\Pi_{P,Q} (R) - \Pi_{P,Q} (S)))$

- ☐ I and II
- ☐ I and III
- ☒ I, III and IV
- ☐ I, II and III

Seçimi temizle

Which constraint involves only on the key attribute(s)?

- ☐ Semantic Integrity
- ☐ Entity Integrity
- ☒ Referential Integrity
- ☐ None
- ☐ Domain Constraint

Seçimi temizle

In an RDBMS relationships between tables are created by using

- ☐ Composite keys
- ☒ Foreign keys
- ☐ Candidate keys
- ☐ Alternate keys

Seçimi temizle

If every non-key attribute is functionally dependent on the primary key, then the relation is in _____ .

- ☐ Third normal form
- ☒ Second normal form
- ☐ Fourth normal form
- ☐ First normal form

Seçimi temizle

Which of the following is a fundamental operation in relational algebra?

- ☒ None of the above
- ☐ Assignment
- ☐ Natural Join
- ☐ Set intersection

Seçimi temizle

Given the basic ER and relational models, which of the following is incorrect?

- ☐ In a row of a relational table, an attribute can have exactly one value or a NULL value
- ☒ In a row of a relational table, an attribute can have more than one value
- ☐ An attribute of an entity can be composite
- ☐ An attributes of an entity can have more than one value

Seçimi temizle

Geri

Gönder

Sayfa 3 / 3

Google Formlar üzerinden daha hızlı gönderim için.

Bu içerik Google tarafından oluşturulmuş ve yayımlanmıştır. [Közüye Kullanımı](#) - [Hizmet Sağlama](#) - [Gizlilik Politikası](#)

Google Formlar